

defumat

Plane-wave DFT in Python and JAX, validated against Quantum ESPRESSO

2026-08-26

A ground-up reimplementation of `pw.x` — *not* a wrapper, not a binding.
Every number below is measured against Quantum ESPRESSO on the same input.

What it is, and why JAX

Two reasons, and both constrain the code:

1. **Autodifferentiation.** Forces, stress, phonons, dielectric response, Raman tensors come from *differentiating the code*, not from separately hand-derived expressions.
2. **GPU.** The same source runs on an accelerator unchanged.

The consequence that matters:

QE's six hand-derived force terms, its stress terms, its `force_hub` — these exist here as *tests*, not as the implementation. One gradient replaces them.

Written down once, differentiated thereafter

energy	written
V_{xc}	<code>jax.grad</code>
force	<code>jax.grad</code>
stress	<code>jax.grad</code>
dyn. matrix	jvp of the force
$d\epsilon/d\tau$ (Raman)	jvp of F_{ij}

Third derivatives of E_{xc} fall out of the same kernel, so a **GGA Raman tensor** works where `ph.x` stops.

Ground state

- ▶ SCF, bands, DOS / PDOS, tetrahedra
- ▶ norm-conserving, **ultrasoft**, **PAW**
- ▶ LDA, PBE, revPBE, PBEsol; **meta-GGA** (TB09)
- ▶ metals: all smearings + tetrahedron occupations
- ▶ collinear spin, **noncollinear**, **spin-orbit**
- ▶ **DFT+U**, magnetic fields, constrained moments
- ▶ Grimme D2 dispersion

Structure

- ▶ forces, stress, BFGS relax, **variable-cell** relax

Response (all by autodiff)

- ▶ dielectric tensor, Born charges
- ▶ **phonons** at Γ (insulators and metals)
- ▶ elastic constants, **electrostriction**
- ▶ **Raman + infrared spectra**

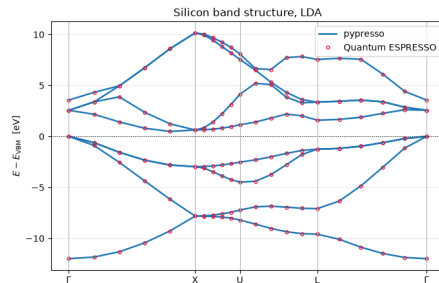
No Quantum ESPRESSO counterpart

- ▶ **spin spirals** (generalized Bloch theorem)
- ▶ **spiral wavevector relaxation**, dE/dq
- ▶ **Chern numbers**, Z_2 invariants
- ▶ rank- n tensor symmetrisation

Refused *by name* rather than silently approximated: PAW Born charges, $\chi^{(2)}$, phonons at $q \neq 0$, EXX.

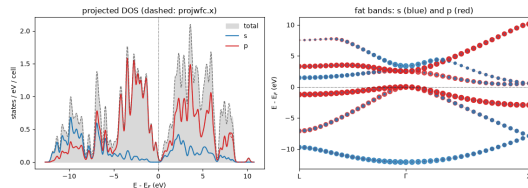
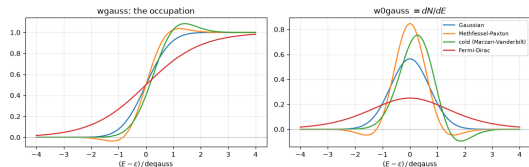
Agreement with Quantum ESPRESSO: the ground state

quantity	agreement
total energy, Si (NC, LDA)	$\sim 1\text{e-}9$ Ry
ultrasoft / PAW	$\leq 3\text{e-}9$ Ry
PBE family, all three kinds	$\leq 6\text{e-}9$ Ry
metals, every smearing	$2.5\text{e-}8$ Ry
collinear spin (Ni)	$1.2\text{e-}9$ Ry
spin-orbit (Pt)	$\leq 1.3\text{e-}8$ Ry
noncollinear magnetism (Fe)	$2.8\text{e-}9$ Ry
DFT+U (FeO, Ni)	$\leq 6.7\text{e-}9$ Ry
band structure	0.0002 eV
forces	$\leq 2\text{e-}5$ Ry/bohr
stress (13 cases)	$\leq 2.7\text{e-}7$ Ry/bohr ³



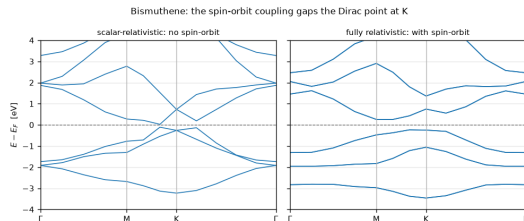
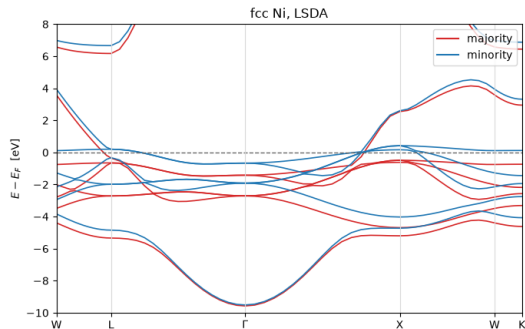
Silicon bands against the committed Quantum ESPRESSO benchmark.

Densities of states, projected and total



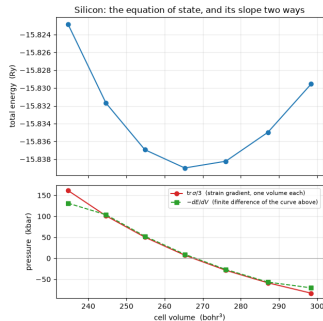
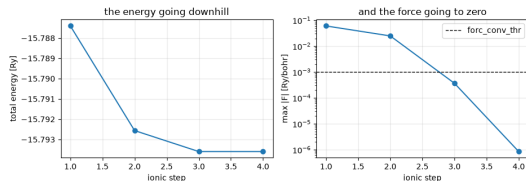
Both smearing and tetrahedron families, the latter also as an occupation scheme inside the SCF. **PDOS** is $\langle \phi | S | \psi \rangle$ on Löwdin-orthogonalised orbitals, matching projwfc.x to 6.9e-4 on a projection and 4.7e-5 on a Löwdin charge; the spilling parameter reproduces all four printed decimals.

Magnetism: collinear, noncollinear, spin-orbit



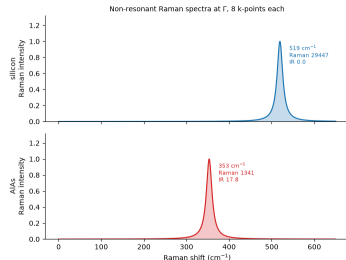
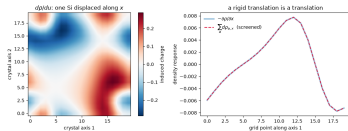
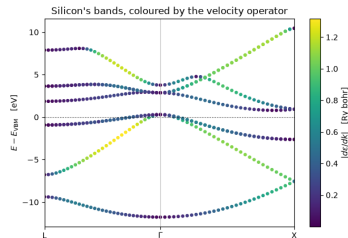
Three spin numbers are kept apart — `nspin` (regime), `npol` (spinor components of a *wavefunction*), `nspin_mag` (components of a *density*) — which is what stops a nonmagnetic spin-orbit run allocating a magnetization it does not have. Nickel's moment: 0.7280 against the 0.73 Quantum ESPRESSO prints.

Forces, stress and relaxation — one gradient, not six terms



The force is `jax.grad` of the total energy at frozen wavefunctions: Hellmann-Feynman, Pulay *and* the augmentation charge's own derivative all fall out of one gradient. QE's six terms are implemented beside it as a **cross-check** — which found the augmentation force's sign. Relaxed geometries reproduce Quantum ESPRESSO to 1e-6 bohr.

Response: dielectric, phonons, Raman



Sternheimer, not a sum over states — no empty bands, no division by $\epsilon_n - \epsilon_m$. Silicon's ϵ_∞ matches `ph.x` to $\leq 1.2 \times 10^{-4}$ on NC, ultrasoft *and* PAW; its optical mode is 510.10 cm^{-1} against 510.15; aluminium's metal phonons agree to 0.002 cm^{-1} . Silicon's T_{2g} Raman line: 519.2 cm^{-1} against an experimental 520 — Raman-active and infrared-silent, which is a symmetry statement rather than a fit.

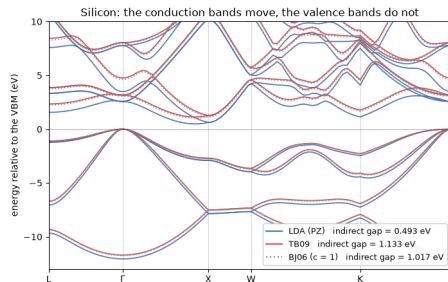
Band gaps: the Tran-Blaha meta-GGA

A functional whose *potential* is written down and whose energy does not exist. `input_dft = 'tb09'` gives the modified Becke-Johnson meta-GGA, which fixes the gap without the cost of a hybrid.

	LDA	TB09	experiment
silicon	0.49	1.13	1.17
diamond	3.89	4.43	5.5

Validated against *analytic* limits rather than another code's floating point: the hydrogen atom, where Becke-Roussel is the exact Slater potential of the 1s orbital to $1e-6$ and E_x is exactly $-5/16$ Ha, and the uniform gas, where the model reproduces v_x^{LDA} to $6e-4$.

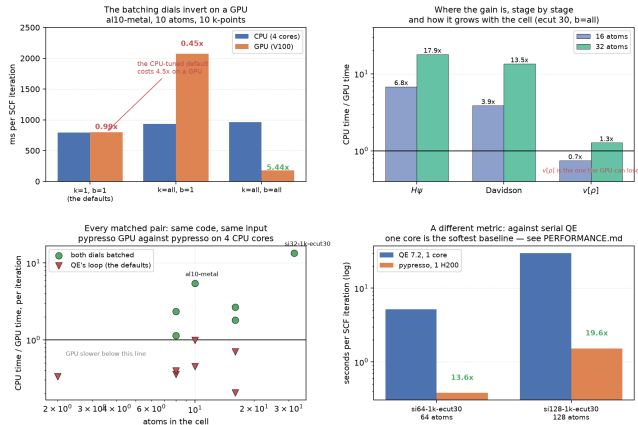
PAW is what makes it work: its one-centre τ recovers $c = 1.107$ against an all-electron 1.12, where norm-conserving silicon measures 1.000.



The consequences are enforced rather than documented: the total energy is **not variational**, so forces, stress, phonons and response are **refused by name** for this functional.

GPU: nothing was ported — what was missing was evidence

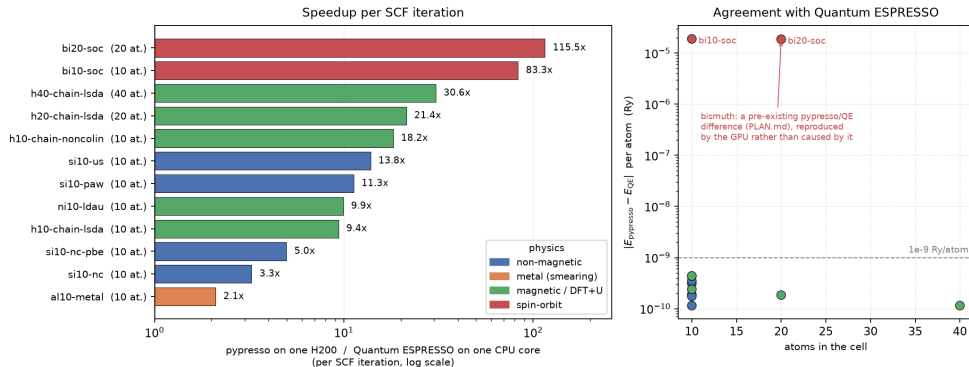
pypresso on a GPU — every CPU comparison measured, 2026-08-25/26



JAX already emitted GPU code from this source. **The finding is that the two batching dials invert:** tuned for a CPU cache they cost **4.5x** on a GPU, and the plausible half-measure ($k=all$, $b=1$) is the *worst* setting.

GPU across the physics — 12 cases, 10–40 atoms

pypresso on a GPU against Quantum ESPRESSO on one CPU core — 12 cases, conv_thr = 1e-10



Speedup tracks **work per k-point**: 2x on a cheap metal, 9–21x with an augmentation charge or a magnetization, **83–115x on spin-orbit**, where serial Quantum ESPRESSO spends 22–105 seconds per iteration on two-component spinors. Energies agree to $\sim 1e-9$ Ry — reproducing the CPU agreements already on record, including the one known bismuth discrepancy.

Done

- ▶ SCF $\rightarrow$ bands $\rightarrow$ DOS, with US/PAW, GGA, all spin regimes
- ▶ forces, stress, relaxation, variable cell
- ▶ linear response, phonons, Raman/IR, electrostriction
- ▶ topological invariants, spin spirals (no Quantum ESPRESSO counterpart)
- ▶ **first contact with a GPU**, validated across the physics

Next

- ▶ phonons at $q \neq 0$; $q2r/matdyn$ dispersions
- ▶ $\chi^{(2)}$ and the electro-optic tensor
- ▶ ultrasoft dynamical matrix, PAW Born charges
- ▶ GPU: k-axis sharding, multi-GPU, the subspace algebra

Two habits do the work. Every quantity is written down *once* and differentiated thereafter, with QE's hand-derived expressions kept as tests — so the two implementations share no machinery and disagreeing is informative. And anything not implemented is **refused by name** rather than silently approximated.